Hashing, passwords and authentication



Cyber security · University · Lecture 3

In this lecture

- State hash properties.
- Store passwords properly.
- Choose a second factor.

Remember

- Fast hashes are wrong for passwords.
- Salt every user.
- Prefer phishing-resistant MFA.

Crack a fake table

1. Warm-up (5 min): How fast?
2. Teach (10 min): Hash properties.
3. Crack (25 min): Three tables.
4. Decide (15 min): A policy.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. How many accounts fell in each table?
2. Why do salts matter even with slow hashes?
3. Which factor would you require for staff?

Exit ticket

- Write the storage line you would use for one password.